

Thinking less, trusting more: genAI's impacts on students' cognitive habits

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ABSTRACT

Context: Many students now use generative AI (genAI) in their coursework, yet its effects on their intellectual development remain poorly understood. While prior work has investigated students' cognitive offloading during episodic interactions, it remains unclear whether using genAI *routinely* is tied to more fundamental shifts in students' thinking habits.

Objectives: To explore this possibility, we investigate (RQ1-How): how students' trust in and routine use of genAI relate to their cognitive engagement—specifically, reflection, the need for understanding, and critical thinking in STEM coursework. Further, we investigate (RQ2-Who): for which students these cognitive disengagement patterns are strongest.

Methods: We drew on dual-process theory, cognitive offloading, and the automation bias literature to develop a statistical model explaining how and to what extent students' trust-driven routine use of genAI was associated with their cognitive engagement habits in STEM coursework, and how these associations differed across students' diverse cognitive styles. We empirically evaluated this model using Partial Least Squares Structural Equation Modeling on survey data from 299 STEM students across five North American universities.

Results: Students who trusted and routinely used genAI reported significantly lower cognitive engagement. Unexpectedly, these associations were strongest among students with traits often celebrated in STEM, i.e., with higher technophilic motivations, risk tolerance, and computer self-efficacy. Interestingly, these associations did not significantly differ by students' prior genAI experience or academic level.

Conclusion: Our findings suggest the potential for a *cognitive debt cycle* in which routine genAI use and weakened intellectual habits reinforce one another. This poses critical challenges for curricula and genAI system design, requiring interventions that actively support cognitive engagement.

1. Introduction

“People are neither fully rational nor completely selfish — and they are often lazy thinkers.” — Daniel Kahneman

Human cognition evolved to be lazy and for good reason. Our ancestors who conserved mental energy for genuine threats survived; those who exhaustively analyzed every decision did not. This “law of least effort” pervades human behavior [2–4]. We outsource memory to shopping lists, arithmetic to calculators, and navigation to GPS, trading internal cognitive work for external support. Generative AI (genAI) tools (e.g., ChatGPT, Claude, Copilot) tend to amplify this tendency by delivering fluent, confident support on demand. As a result, interacting with genAI involves less “thinking by doing” and more “choosing from outputs” [5]. In many domains, this reallocation saves time, improves

productivity, and creates a compelling sense of progress [6].

But what works for productivity can be problematic for learning with unrestricted use. In STEM education, for example, genAI may remove the desirable difficulties students need [7]: slow, effortful, and often uncomfortable cognitive work that builds intuition and transferable skills [8]. Students may still produce equal (or even better) quality work yet learn less [9]. Recent studies have shown that genAI assistance usage reduced depth of inquiry, evidence use, and self-regulatory behaviors in students [10–12,7,13,14]. Of course, these harms are not inevitable; genAI use can help learning, for example, as tutors when designed around pedagogical best practices [15,16].

Whether genAI supports or undermines learning thus hinges on how critically users engage with its output. Automation bias research shows that when systems appear competent, users trust outputs with limited scrutiny—even when those outputs conflict with their own knowledge

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or judgment [17,18]. In human-AI interaction (HAI) contexts, this tendency can manifest as over-reliance [19,20], with users incorporating AI suggestions without much critical evaluation [21,5]. These effects are amplified in settings that reward immediate efficiency, including professional work [6,5], and increasingly academic contexts [9], where genAI use is becoming routine.

Consider these two scenarios:

Scenario A (episodic): A student uses genAI to bypass a single difficult problem, showing momentary *overreliance* and *reduced cognitive effort*.

Scenario B (routine): Through trust-driven routine genAI use, that student develops a general unwillingness to *engage* in reflection, understanding, and critical thinking *across coursework*—fundamentally recalibrating what cognitive effort investment feels “worth it”.

Prior work has largely focused on Scenario A [10–12,7,13,22,14]. In contrast, this paper focuses on Scenario B—how and to what extent *routine* use of genAI relates to students’ cognitive engagement habits. If such effects exist, are STEM students becoming adept at querying intelligence while exercising less of their own? Would they then enter professional work with fewer opportunities to develop the cognitive habits that prior generations built through practice? As a first step towards answering these long-term questions, we ask: **RQ1-How: To what extent are students’ trust in and routine use of genAI associated with their cognitive engagement habits in STEM coursework?** Drawing on decades of research in educational psychology and learning sciences [23–27], we conceptualize cognitive engagement as three intellectual habits: (1) *Reflection* (the metacognitive monitoring and evaluation of one’s understanding) [28,26,29], (2) *Need for understanding* (the intrinsic motivation to seek conceptual coherence rather than surface-level correctness) [30,31,24], and (3) *Critical thinking* (the analytic evaluation of evidence, assumptions, and alternatives) [32–34] (detailed in Sec. 3.1). Foundational education research shows that these habits foster durable learning and skill development [35,36]; RQ1-How investigates how/how much all three are related to students’ trust in and routine offloading of cognitive work to genAI.

However, these effects are unlikely to be the same for all individuals. Prior work shows that differences in individuals’ cognitive styles [37, 38] substantively shape the ways they use technology [39–41]. (We expand upon the specific cognitive styles in Sec. 3.2). Further, recent HAI studies suggest that these differences, alongside trust, influence how individuals adopt genAI [42]. Thus, we ask: **RQ2-Who: Who (i.e., students with which cognitive styles) are particularly prone to the cognitive-disengagement pattern associated with routine genAI use?**

Our study makes three contributions: (1) a theoretically grounded statistical model of (for RQ1-How) how students’ trust-driven routine use of genAI relates to their cognitive engagement habits in coursework, and (for RQ2-Who) for whom these associations are strongest. We then empirically evaluate this model using Partial Least Squares–Structural Equation Modeling (PLS-SEM) on survey data from 299 STEM students across five North American universities. This model and its empirical evaluation also provide (2) a psychometrically validated survey instrument for measuring these constructs in student–genAI interaction contexts. Finally, we derive (3) implications and open challenges for educators and researchers seeking to support diverse individuals’ cognitive engagement in genAI-augmented STEM education.

2. Related work

The past few years have seen a rapid adoption of genAI in educational settings, prompting growing interest in how these systems shape students’ learning processes and outcomes. In response, a substantial body of work examined the effects of genAI use on cognition, learning, and task performance across educational levels and domains [43,12,9,7,

44,22].

Much of this work reports a consistent pattern: genAI supported short-term efficiency gains (e.g., faster ideation, improved writing fluency, note-taking, and problem-solving) while coinciding with weaker higher-order learning outcomes (e.g., creativity, metacognition, self-regulation) [9,13,14,22]. For instance, in secondary education, Kreijkes et al. [14] found that students who relied on AI-generated notes perceived comprehension tasks as easier and expended less effort, yet demonstrated weaker retention than peers who generated their own notes.

Related findings were reported in higher education contexts. Several studies observed that while genAI use reduced perceived task difficulty and mental effort, it was also associated with diminished depth of inquiry, evidence use, reasoning quality, and self-regulatory behaviors [10,11,7]. For example, Barcaui [7] found that undergraduates given unrestricted ChatGPT access for learning performed substantially worse on assessments administered weeks later, attributing this decline to the bypassing of “desirable difficulties” (e.g., self-quizzing, note-taking) essential for consolidating knowledge. Similarly, Fan et al. [11] reported that students revising essays with ChatGPT showed reduced motivation to rely on source material, exhibiting “metacognitive laziness” when progress felt effortless.

Other work examined how these effects varied with how genAI was incorporated into learning activities [16,45]. In one such study, Choudhuri et al. [16] found that software engineering students benefited when genAI was used to clarify concepts or resolve impasses following independent effort, but experienced breakdowns when it was used prematurely—during initial learning or for complex implementations without sufficient foundational understanding.

These conditions matter more broadly: outcomes depend on how genAI is embedded in learning. For instance, an AI tutor designed around pedagogical best practices was shown to outperform in-class active learning in a randomized study, with students learning more in less time [15]. Further, meta-analytic work reports positive effects of ChatGPT on learning performance, moderated by course type, learning approach, and how the tool is used (e.g., scaffolded tutoring, instant feedback) [46]. Interaction style matters as well: students who read, re-prompted, and edited outputs outperformed those who copy-pasted them [47]. Synthesizing this evidence, a recent review suggests that learning gains depend on student readiness, AI literacy, and use as a supplement rather than a substitute [9]. This motivates our focus on routine, self-directed use in everyday coursework, the mode in which such structure is typically absent.

Beyond immediate task performance, several studies suggested that genAI use could create an “illusion of competence” and a sense of progress that outpaced actual understanding [48]. For example, Prather and colleagues [49,50] found that students using genAI tools reported higher confidence and completed tasks more quickly (sometimes producing higher-quality artifacts), yet faced metacognitive difficulties (e.g., challenges in debugging, explaining their own solutions, or adapting when AI-generated suggestions failed) [49,44].

Complementing these behavioral findings, Kosmyrna et al. [12] examined neurophysiological correlates of cognitive offloading. In a controlled lab study, college students writing with genAI assistance showed reduced activation in brain networks associated with creativity and cognitive control, alongside weaker subsequent memory recall and a lower sense of ownership over their written contributions.

Notably, these dynamics extend beyond classrooms. Studies of genAI use among knowledge workers reported similar patterns of cognitive offloading [5,6]. Trust in AI appeared to further amplify these effects: when systems were perceived as competent, users were more likely to accept outputs with limited scrutiny [21,51]. For example, Lee et al. [5] found that workers with higher confidence in genAI engaged less in creativity and critical thinking during tasks, shifting from deliberate reasoning toward passive evaluation of AI-generated content. Broader population studies likewise linked frequent reliance on genAI with

increased offloading, work-slop, and overconfidence [43,52].

Overall, prior work has largely focused on episodic cognitive offloading and its immediate effects on learning and task performance. What remains underexplored is how routine genAI use during formative education reshapes students' enduring habits of cognitive engagement. Our study addresses this gap by examining how students' trust in and routine use of genAI relate to students' reflection, need for understanding, and critical thinking in STEM coursework (RQ1-How). Further, prior work has shown that variations in individuals' cognitive styles affect technology adoption [40,42], and that individual factors shape how students engage with genAI [53,48,9]. To the best of our knowledge, however, there is little work modeling how these individual differences relate to students' cognitive engagement under routine genAI use, which we examine in RQ2-Who.

3. Foundations and model development

We draw on foundational work in HCI and cognitive psychology to develop a theoretical model explaining two pathways of influence (see Fig. 1): (1) how students' trust in and routine genAI usage relate to their cognitive engagement habits (RQ1-How), while (2) accounting for variations in their cognitive styles (RQ2-Who).

3.1. Cognitive engagement

Consider what happens when you encounter a confusing passage in a textbook. One response is to highlight it and move on, trusting that re-reading will somehow clarify things later. Another response is to stop, notice the confusion, and deliberately work to resolve it: "Wait—how does this claim follow from the previous one? What am I missing?" The second response reflects *cognitive engagement*. It involves making sense of

ideas, recognizing gaps in your own understanding, and investing effort to close them, rather than simply collecting information or reproducing facts [54,55].

Educational psychology has long characterized cognitive engagement habits across three core intellectual di-mensions: (a) *Reflection*, (b) *Need for understanding*, and (c) *Critical thinking* [23–27]. Learners who cultivate these habits are more likely to develop durable understanding and transferable skills than those who rely on more superficial strategies [24–26]. In this study, we therefore conceptualize cognitive engagement along these dimensions. Below, we define each dimension and describe how it is operationalized in our model.

Reflection is the metacognitive habit of monitoring one's own understanding, evaluating assumptions and strategies, and recognizing gaps in thinking [28]. For example, when you read that ocean temperatures are rising by 0.13°C per decade, do you catch yourself wondering whether that's fast or slow, and what it implies? Education reformer John Dewey described reflection as "overcoming the inertia that inclines one to accept suggestions at their face value" [23]. Research shows that reflective learners are better positioned to catch their own errors, recognize when they're stuck, and adjust learning strategies adaptively [28,26,29].

Reflection is a latent psychological construct (subjective, intangible, and not directly observable) and is therefore captured through validated self-report questionnaires [56,24,26]. In our study, we adopt Biggs' validated framework [24], which conceptualizes *Reflection* as a combination of *Reflective Motives* (why learners reflect) and *Reflective Strategies* (how they do so). Consistent with this framework, we model *Reflection* as a higher-order construct (HOC) composed of these two lower-order constructs (LOCs) [57], represented as circles in Fig. 1. Each LOC is itself latent and measured through multiple self-report items (questions); in our model, *Reflective Motives* and *Reflective Strategies* are measured

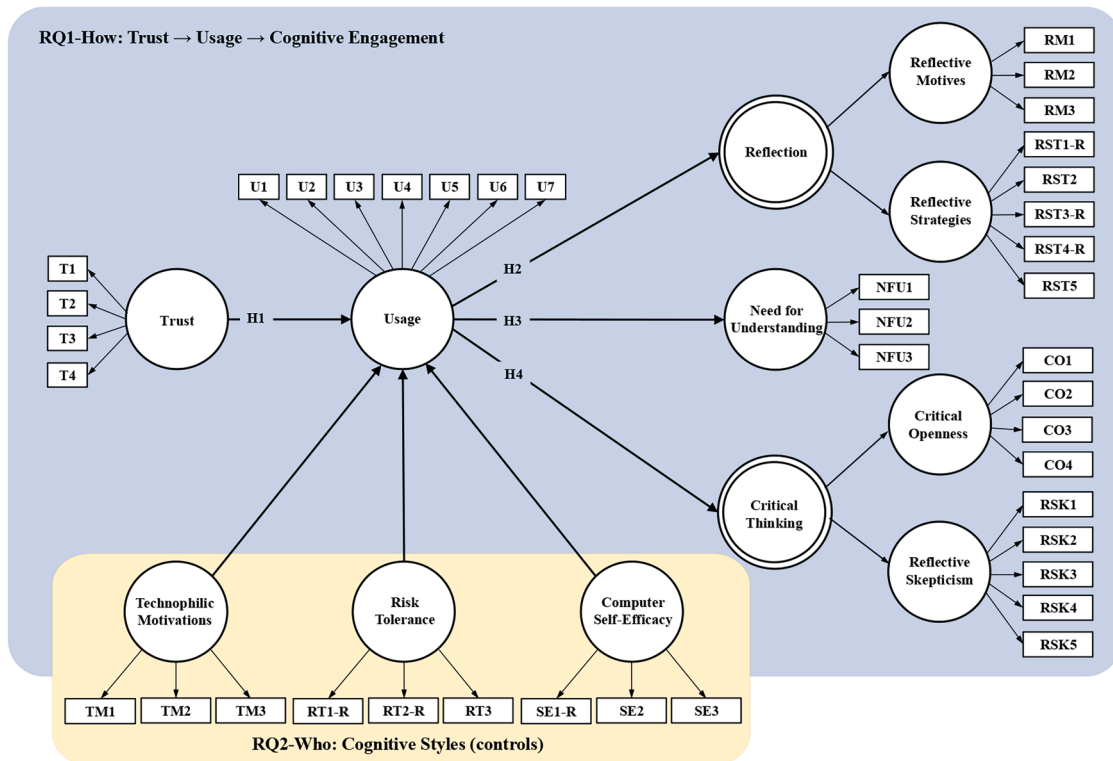


Fig. 1. Proposed theoretical model and measurement specification. The blue region (RQ1-How) models how *Trust* and routine *Usage* relate to cognitive engagement: (a) *Reflection*, (b) *Need for Understanding*, and (c) *Critical Thinking*. The yellow region (RQ2-Who) controls for cognitive styles (*Technophilic Motivations*, *Risk Tolerance*, *Computer Self-Efficacy*). Latent constructs (circles) are reflectively measured by survey items (squares) [57]; higher-order constructs (double circles) are reflectively measured via their lower-order constructs (e.g., *Reflection* is measured via *Reflective Motives* & *Strategies*). Reverse-coded items carry the “-R” suffix (e.g., RT1-R). Directed arrows represent hypothesized relationships (H1–H4) and cognitive-style controls. Item wordings (questionnaire) are in [1].

using three and five items respectively (denoted as squares in Fig. 1).

Need for Understanding captures a learner's intrinsic drive to make sense of things [30,31]. Some learners are satisfied once they can execute a procedure correctly. Others experience cognitive discomfort until they grasp *why* it works. This is not stubbornness or perfectionism—it is a stable difference in what feels intellectually satisfying [31]. Learners high in this trait habitually press for explanations and expend cognitive effort to construct meaning, not because they're told to, but because surface-level answers feel incomplete. Unsurprisingly, research shows they learn more effectively: their constructive sensemaking processes foster effective self-regulation that produces durable learning [8,36]. Following prior work [24], we model Need for Understanding as a unidimensional construct, measured using items adapted from Kember's validated instrument [26] (Fig. 1).

Critical Thinking is the analytic work learners do in inspecting assumptions, weighing evidence, and considering alternatives [32–34]. For example, when a solution fails, some may fixate on the most obvious cause. A critical thinker, on the other hand, considers multiple hypotheses: *Is the error due to my inputs, or an unstated assumption in the problem? What evidence would distinguish between these possibilities?*

Critical thinking has been assessed in multiple ways spanning self-report scales, performance tasks, observations, and expert evaluations [32,58,26]. The APA Delphi consensus [32] characterizes critical thinking through two complementary dimensions: (1) *Critical Openness*, the willingness to entertain alternative perspectives and/or competing ideas, and interpret information from multiple angles; and (2) *Reflective Skepticism*, the discipline of questioning evidence and scrutinizing claims. Effective critical thinking requires both [59]: openness without skepticism produces gullibility; skepticism without openness produces rigidity. Accordingly, we model Critical Thinking as an HOC composed of these two LOCs (Fig. 1) [32,34,60], using a validated instrument from [33].

We adopt this definition over alternatives such as Bloom's taxonomy [61] or the Paul-Elder framework [62] for three reasons: First, the Delphi-report formulation is grounded in empirical consensus among multiple experts in psychology and education. Second, its parsimonious two-factor structure (unlike the Paul-Elder framework [62], which spans eight “elements of thought”, ten “intellectual standards”, and eight “intellectual virtues”) aligns with our survey design and structural modeling goals. Third, it lowers measurement complexity and multicollinearity (highly fragmented constructs become unwieldy [57]), yielding a model that is more straightforward to estimate and interpret.

3.2. Model development: how trust, usage, and cognitive styles relate to cognitive engagement

To understand how genAI tools relate to cognitive engagement, we must first understand something fundamental about how human cognition works, or more precisely, how it avoids working. These theories form the basis of our hypotheses:

3.2.1. The laziness of thought (dual process and cognitive offloading theory)

Our minds did not evolve to think carefully about everything. That would be metabolically ruinous and strategically foolish [4,2]. According to Kahneman's dual-process theory [4], human cognition operates through two systems: System 1 (“*fast thinking*”) is intuitive, automatic, and low-effort, drawing rapidly on associations and patterns. For example, it is what delivers the answer “ $2 + 2 = ?$ ” without deliberation, or completes the phrase “*bread and...*”. System 2 (“*slow thinking*”), on the other hand, is deliberate, controlled, and effortful. It is invoked when, for example, you calculate 17×24 , evaluate a complex argument, or consciously monitor your behavior in a social situation.

System 2 is lazy. It avoids effort whenever System 1 can provide a plausible solution. This is not a bug; it is an evolutionary feature. Our ancestors who conserved cognitive resources for genuine threats

survived; those who exhaustively analyzed every decision were out-competed. The result is what psychologists call the “*law of least effort*” [4]—a pervasive tendency to accept the path of least cognitive resistance.

This explains why cognitive offloading is so natural [2]. When external aids can shoulder cognitive demand, we instinctively let them—*why waste cognitive resources solving problems that have already been solved?* In learning contexts, however, it can cause students to skip over mental processes that are integral to the formation of knowledge. In the past, we have seen the “*Google effect*” [3]: people became less likely to remember information they believed would be available online. Recent HAI work shows similar offloading patterns with genAI, where users lean on AI-generated suggestions at the expense of their own thinking [5,12].

3.2.2. Automation bias

The tendency of cognitive offloading is compounded by trust [17, 18]. When automated systems appear competent, users may accept outputs with limited scrutiny—even when those outputs conflict with their own knowledge or judgment [17,18]. In HAI contexts, this manifests as over-reliance: users defer judgment to AI, rapidly integrating its suggestions while failing to consider the likelihood or occurrence of errors/problems [21,20,63].

3.2.3. From laziness of thought and automation bias to RQ1-How's hypotheses

Drawing on dual-process, cognitive offloading, and automation bias theories, we posit relationships linking students' trust in and routine use of genAI with their cognitive engagement habits in coursework.

First, we posit that *trust is positively associated with routine usage*. We adopt a widely used definition of trust in AI: “*the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability*” [63–66]. Across automation and HAI research, trust has been shown to predict adoption [63,67,68]. Students who view genAI as reliable are likely to consult (and defer to) it more frequently for their coursework. Accordingly:

- H1: Trust is positively associated with routine genAI usage.

Second, we posit that *routine usage is negatively associated with cognitive engagement habits*. Dual-process accounts suggest one possible mechanism: repeated offloading could reinforce a fast System-1 response, “*problem*

→ *offload to genAI*”, that could displace deliberate cognitive engagement. Accordingly, we hypothesize:

- H2a. Routine genAI usage is negatively associated with Reflection.
- H3a. Routine genAI usage is negatively associated with Need for Understanding.
- H4a. Routine genAI usage is negatively associated with Critical Thinking.

Third, we posit that *trust relates to cognitive engagement habits through routine usage*. Where trust coincides with routine use, that repeated offloading could reinforce the System-1 response described above. Accordingly:

- H2b. Trust is transitively associated with lower Reflection through increased routine usage.
- H3b. Trust is transitively associated with lower Need for Understanding through increased routine usage.
- H4b. Trust is transitively associated with lower Critical Thinking through increased routine usage.

Following prior work, we adapt validated instruments to measure students' trust (4-item TXAI instrument [69,68]) and routine use of

genAI (academic genAI use instrument [70]).

3.2.4. Beyond how's to who's (RQ2-Who)

Individuals are not identical in their propensity to use tools. Instead, they differ in their *cognitive styles*, i.e., *the way they perceive and process information, problem solve, and work with technology* [37–39], which affects their technology adoption [40,41]. Recent HAI work shows that certain cognitive styles: *technophilic motivations* (intrinsic interest in engaging with technology), *risk tolerance* (comfort with uncertainty), and *computer self-efficacy* (confidence in using technology) significantly predict genAI adoption [42].

Thus, to avoid conflating these stable traits with the effects of trust modeled in RQ1-How, we include technophilic motivations, computer self-efficacy, and risk tolerance as **control variables** on genAI usage: they help explain baseline differences in *who* is most likely to heavily adopt genAI [68], and in turn, *who* may show the strongest disengagement patterns. We capture these cognitive styles using the validated GenderMag facet survey [71], following prior work [68].

Finally, given evidence that academic level and prior genAI experience can influence usage [72,16], we performed **group analyses** to test whether our model relationships varied across these groups.

4. Method

4.1. Survey design

To address our RQs, we conducted a large-scale survey with STEM students across five North American institutions. We followed Kitchenham's guidelines for designing empirical surveys [73], and drew on established frameworks from educational psychology and Human–AI Interaction studies to measure our constructs (Table 1). Each construct in our model (circles in Fig. 1) represents a latent variable, i.e., a variable not directly observable but inferred through validated indicators (e. g., survey questions, squares in Fig. 1). We adapted existing questions to the genAI context, where applicable. For example, a risk-tolerance item reading “I avoid advanced [technology] features or options” was revised to “I avoid advanced genAI features or options” (see supplemental [1]).

Our survey comprised three sections:

- (1) **Cognitive engagement.** After providing informed consent, participants reported their cognitive engagement habits in coursework, including their reflection (reflective motives and strategies), need for understanding, and critical thinking (critical openness and reflective skepticism), captured using validated instruments listed in Table 1.
- (2) **Trust, usage, and cognitive styles.** Participants reported their trust in and usage of genAI tools in coursework, alongside their cognitive styles, including risk tolerance, technophilic motivations, and computer self-efficacy (Table 1).
- (3) **Background and genAI experience.** Finally, participants rated their prior experience with genAI, academic level, and (optionally) their major, age, and gender (see supplemental material [1]). During the pilot phase, participants were also invited to share open-ended feedback on the survey.

Table 1
Constructs in the theoretical model and validated instruments used to measure them.

Construct	Instrument
Trust	TXAI instrument [69,68]
Usage	Academic genAI usage instrument [70]
Reflection	Biggs' instrument [24]
Need for Understanding	Kember's instrument [26]
Critical Thinking	CTDS instrument [33]
Cognitive Styles	GenderMag facet survey [71,68]

We administered the survey in Qualtrics [74]. All closed-ended questions used a five-point Likert scale (1 = *strongly disagree*, 5 = *strongly agree*) with a neutral option and a sixth “I'm not sure” option to distinguish ignorance from indifference [75]. The survey took 7–10 minutes to complete. We included attention-check questions and randomized item order within blocks to reduce response bias. The survey questionnaire is available in the supplemental material [1].

Ethics approval. The study was approved by the Institutional Review Board (IRB) at Oregon State University in accordance with the Declaration of Helsinki. All participants provided electronic informed consent before participating. **Sandbox and pilot.** We sandboxed the survey one-on-one with psychology/HCI/CS-education researchers ($n = 9$) and STEM instructors ($n = 4$) to assess interpretability, clarity, and realism; revising ambiguous question phrasing based on their feedback. We then piloted it with 67 STEM undergraduates, confirming item clarity and construct validity. We made minor wording edits and excluded pilot responses from the final analysis.

Validation. We psychometrically validated our survey instrument following [57] (see details in Sec. 4.3).

4.2. Data collection

Distribution. The survey was available for three months (April to June 2025). We recruited STEM instructors at our institution and via education conferences and networks, who then distributed the survey to students in their courses at their own discretion. We asked instructors to encourage participation and, where appropriate, to offer it as an optional, non-graded exercise. Participation was voluntary, uncompensated, and anonymous, and instructors could neither view individual responses nor identify who had participated. We did not have access to class rosters, so the number of students who received the invitation is unknown and we cannot report a response rate. We contacted instructors rather than students because this broadened our reach across institutions and course types and aligned with our IRB guidelines.

Sample size determination. We determined the minimum sample size for validating the model via power analysis using the G*Power tool [76]. We specified an F-test with multiple linear regression to detect even small effect sizes ($f^2 = 0.10$) at a significance level of ($\alpha = 0.05$) with power = 0.95, and assuming at most four predictors per construct.¹ The analysis suggested a minimum sample size of 191.

Responses. We received a total of 392 responses. We removed patterned responses ($n = 20$), partial submissions ($n = 39$) and those that failed attention checks ($n = 30$). We retained **299 valid responses** after filtration, which exceeded the *a priori* sample size requirement of 191. Respondents reported a wide distribution of genAI experience, usage frequencies, academic level, age, and gender, consistent with prior studies [5,77]. A summary of respondent demographics is available in the supplemental material [1].

Missing data. We treated “I'm not sure” selections as missing data. Across the retained sample, 1.1% of item responses were missing. Missingness was lowest for Usage (0.2%), followed by reflection 1.1%, trust was 1.6%, need for understanding 1.7%, and critical thinking 2.3%. The cognitive style controls carried the highest rates: risk tolerance (2.4%), computer self-efficacy (2.5%), and technophilic motivations (2.8%). No single item exceeded 6.1% missing, and 94.4% of retained respondents answered every model item.

We did not impute missing values given the unproven efficacy of imputation in SEM group contexts [78]. We instead used pairwise deletion under SmartPLS defaults, in which each parameter is estimated from the cases available for the indicators involved. Given the low overall rate (1.1%), this had a limited effect on estimation and 94.4% of

¹ In Fig. 1, the construct with the most incoming arrows is Usage, predicted by (1) Trust, (2) Technophilic Motivations, (3) Risk Tolerance, and (4) Computer Self-Efficacy.

respondents contributed complete data to every path. Little's MCAR test was non-significant ($\chi^2(3072) = 3100.31, p = .356$), giving no evidence that missingness depended on observed responses. We also re-estimated the full model using casewise deletion ($n=281$). All paths retained their sign, significance level, and effect size category, with coefficients differing by at most .004, suggesting our findings are not sensitive to how missing responses were handled.

4.3. Data analysis & survey validation

We analyzed the survey data using *Partial Least Squares–Structural Equation Modeling* (PLS-SEM) to empirically evaluate our theoretical model. PLS-SEM suits exploratory studies like ours because it estimates multiple relationships among constructs while accounting for measurement errors in one comprehensive analysis [77,68]. Importantly, it relaxes multivariate distributional assumptions. Instead, it calculates the statistical significance of path coefficients (i.e., relationships between constructs) through *bootstrapping*: resampling thousands of subsamples (5,000 in our case) to derive inferences [79].

We conducted the analyses using *SmartPLS 4* [80] in two phases. First, we evaluated the *measurement model to validate our survey*: i.e., verify that the items (questions) reliably and validly capture the intended constructs. Second, we evaluated the *structural model* to test the hypothesized relationships among these constructs. Below, we detail the *measurement model* evaluation (Steps 1–5). The *structural model* evaluation and results are detailed in Sec. 5.

Survey validation (measurement model evaluation): We used a two-stage procedure to validate our survey instrument, i.e., assess both lower-order and higher-order reflective constructs (circles and double circles in Fig. 1) following recommended PLS-SEM practices [81,82,79]. The data met standard assumptions for measurement-model analysis: Bartlett's test of sphericity was significant for all constructs ($\chi^2(946) = 7167.278, p < .001$), and the KMO measure of sampling adequacy was 0.901, exceeding the 0.60 threshold [83].

Stage 1: Lower-Order Reflective Construct Evaluation: We first assessed the lower-order reflective constructs (Trust, Usage, Reflective Motives, Reflective Strategies, Need for Understanding, Critical Openness, and Reflective Skepticism, circles in Fig. 1) using tests for (1) convergent validity, (2) internal consistency reliability, (3) discriminant validity, and (4) indicator collinearity [57].

Convergent validity examines whether indicators (questions) intended to measure the same construct share sufficient variance (for reflective constructs, changes in the construct should be reflected in its indicators) [84]. We assessed convergent validity using (a) factor loadings and (b) Average Variance Extracted (AVE) [79].

Factor loadings capture the association between indicators and their constructs (e.g., T1–T4 for *Trust*); values $\geq .60$ suffice for exploratory studies [79]. We removed two weak indicators—U6 of *Usage* and CO1 of *Critical Openness*—that did not adequately reflect their constructs.² All retained indicators exceeded the threshold, with loadings ranging from .60 to .925.

AVE captures the shared variance across a construct's indicators and should exceed .50, indicating the construct explains at least 50% of shared variance among its indicators [79]. All constructs met this criterion (Table 2).

Internal consistency reliability evaluates whether indicators are internally consistent and reliably measure the same construct. We evaluated internal consistency reliability using Cronbach's α and Composite Reliability ($p_{\omega}p_c$) [57]. All constructs met established reliability thresholds (.70–.95) [79] (see Table 2).

Discriminant validity assesses the distinctness of constructs. We used the Heterotrait–Monotrait (HTMT) ratio [85], with adequacy indicated by values $< .90$. HTMT ratios among all constructs met this criterion, supporting

Table 2

Internal consistency reliability, and convergent validity for reflective lower- and higher-order constructs in the model. These metrics reflect whether each construct is measured reliably and captures sufficient shared variance among its indicators. Cronbach's α , Composite Reliability ($CR(p_{\alpha})$, $CR(p_c)$), and Average Variance Extracted (AVE) values indicate internal consistency and convergent validity, with acceptable thresholds of $\geq .7$ (for reliability) and $\geq .5$ (for AVE) respectively.

	Cronbach's α	$CR(p_{\alpha})$	$CR(p_c)$	AVE
Lower-order constructs				
Critical Openness	.746	.802	.735	.524
Need for Understanding	.902	.905	.938	.835
Reflection Motives	.896	.899	.937	.833
Reflective Skepticism	.828	.903	.845	.511
Reflective Strategies	.789	.797	.855	.542
Trust	.818	.831	.883	.655
Usage	.896	.897	.923	.665
Technophilic Motivations	.731	.772	.805	.685
Risk Tolerance	.721	.752	.773	.672
Computer Self-Efficacy	.711	.796	.821	.645
Higher-order constructs				
Critical Thinking	.700	.710	.803	.671
Reflection	.801	.809	.909	.834

Cronbach's α tends to underestimate reliability, whereas composite reliability ($CR(p_c)$) tends to overestimate it. The true reliability typically lies between these two estimates and is effectively captured by $CR(p_{\alpha})$ [57].

2Removing CO1 and U6 also increased AVE from .48 to .52 for *Critical Openness* and from .64 to .67 for *Usage*. discriminant validity. We report HTMT ratios for higher-order constructs and other model constructs in Table 3; ratios for the LOCs of Reflection and Critical Thinking, as well as the Fornell–Larcker criterion and cross-loadings, alongside confirmatory factor analysis, are provided in the supplemental for completeness [1].

Indicator collinearity assessment checks for multicollinearity among reflective indicators. We used the Variance Inflation Factor (VIF) to check for multicollinearity. All indicator VIFs were < 3 , well below the cutoff [79]. **Common method bias.** Since we collected data via a single survey, we tested for Common Method Bias (CMB) [57].

To do so, we first applied Harman's single-factor test on the latent constructs [86]. No single factor explained more than 21.9% variance. An unrotated exploratory factor analysis with a forced single-factor solution explained 26.6% of variance, well below the 50% threshold. Additionally, we used Kock's collinearity check [87]: VIFs for all latent constructs were < 2 , well under the cut-off. These results indicate CMB was not a concern in our study.

Stage 2: Higher-Order Reflective Construct Evaluation: Our model includes two reflective higher-order constructs (HOCs): (1) Reflection and (2) Critical Thinking. Recall, reflection is a motive-strategy pair that acts together (see Sec. 3). Following Biggs et al. [24]'s work, we modeled *Reflection* as an HOC with *Reflective Motives* and *Reflective Strategies* as its lower-order constructs (LOCs). Similarly, following the APA Delphi consensus [32], we modeled *Critical Thinking* as an HOC comprising *Critical Openness* and *Reflective Skepticism*, as in prior work [33]. Collapsing the LOCs into HOCs reduces measurement complexity, aligns the model with theory, and lowers collinearity, yielding a more parsimonious, easier-to-test structure.

HOC assessment. We validated HOCs using the disjoint two-stage approach recommended by Sarstedt and Cheah [88]. We chose this approach because both HOCs in our model (Reflection and Critical Thinking) were endogenous. For endogenous HOCs, the repeated-indicators alternative introduces bias into structural estimates because endogenous HOC's variance is almost fully explained by its own LOCs [81,88]. Additionally, this approach preserved the

Table 3

Heterotrait–Monotrait (HTMT) ratios among model constructs (ranges between 0 and 1). Values < .90 indicate that constructs are sufficiently distinct from each other. HTMT ratios for the LOCs of Reflection (Reflective Motives, Reflective Strategies) and Critical Thinking (Critical Openness, Reflective Skepticism) are reported in [1].

	Critical Thinking	Tech. Motivations	NFU	Reflection	Risk Tolerance	Computer S.E.	Trust
Tech. Motivations	.597						
NFU	.07	.179					
Reflection	.66	.823	.322				
Risk Tolerance	.552	.735	.217	.669			
Computer S.E.	.587	.603	.063	.414	.635		
Trust	.529	.711	.266	.843	.608	.489	
Usage	.587	.661	.228	.806	.786	.526	.725

Tech. Motivations: Technophilic Motivations; NFU: Need for Understanding; Computer S.E.: Computer Self-Efficacy

theoretically specified LOC structure while yielding a parsimonious structural model.

To validate the HOCs, we first estimated latent scores for the lower-order constructs in SmartPLS as linear combinations of indicator scores and outer weights. We then used these scores as indicators for the HOCs: (i) Reflective Motives and Reflective Strategies for *Reflection*, and (ii) Critical Openness and Reflective Skepticism for *Critical Thinking*. Finally, we evaluated the HOCs using the same assessment criteria applied to the LOCs:

Convergent validity. All factor loadings exceeded .70 (Fig. 3), and AVE values were above .50 for both HOCs (Table 2), indicating sufficient explained variance of its indicators (the LOCs).

Internal consistency reliability. Cronbach's α and Composite Reliability values exceeded .70 for both HOCs, indicating sufficient reliability and internal consistency of the constructs (Table 2).

Discriminant validity. HTMT ratios were below .90 for all construct pairs (Table 3), indicating empirical distinctness.

Collinearity. All VIF values were below 2 (see supplemental [1]), indicating no multicollinearity concerns. Together, these stages (1&2) validated our survey instrument: questions reliably capture their intended constructs,

and these constructs satisfy convergent and discriminant validity, enabling us to test structural relationships next (Sec. 5).

4.4. Limitations

As with any empirical study, our work has limitations [89]. All constructs were measured using validated self-report instruments grounded in established theory. Even then, surveys can introduce response biases or misinterpretations. We guarded against these threats by involving instructors in survey design, sandboxing and piloting, randomizing question blocks to reduce order effects, adding attention checks, and screening patterned responses. All constructs met standard thresholds for reliability and validity. Even so, self-report captures self-assessments instead of observed engagement, so our findings are about students' reported cognitive engagement. Establishing effects on enacted cognition will require objective behavioral measures (e.g., performance-based assessments, retention tests) in future work.

As a cross-sectional study [90], we report associations rather than causation; the data do not adjudicate direction, as students already less inclined toward reflection, understanding, or critical thinking may have been more likely to adopt genAI heavily. Self-selection bias remains possible, as students with stronger views may have been more likely to participate. Instructors distributed the survey to their own classes, so respondents were nested within courses and institutions, and instructor expectations could have influenced how students in a given class responded. Voluntary, anonymous, uncompensated, and non-graded participation limits this risk, as does instructors' inability to view responses. Still, we did not collect course or instructor identifiers, so our

model treats responses as independent; future work should collect them and use multilevel designs to estimate how much variation is attributable to classroom context. As with all survey-based research, reported experiences may not align perfectly with in-situ behavior. Nonetheless, given the internal consistency of the results and their triangulation with established theory, we consider this approach sufficient to support the reliability of our conclusions.

Further, a theoretical model cannot capture an exhaustive set of factors. We did not measure baseline academic performance (e.g., GPA) or course difficulty because of IRB restrictions; instead we collected academic seniority (lower- vs. upper-division) as a coarse proxy for the latter. Lower-performing students may rely more heavily on genAI as a coping mechanism, an alternative account that could partially explain the observed associations. Accordingly, our findings should be interpreted as a theoretically grounded starting point; future work should consider longitudinal designs and replication in varied contexts to strengthen causal claims and generalize broadly.

Finally, while the study was designed to be region-agnostic and included students with varied demographics and levels of genAI experience, the sample reflects a primary US-based educational context. Its size and composition are comparable to prior empirical studies in STEM education [91,53], providing a reasonable foundation for theory development. Consistent with this goal, we emphasize theoretical rather than statistical generalizability [92]. Future work should examine transferability across broader populations and contexts.

5. Results

After validating our survey (measurement model), we test our hypotheses by evaluating the structural model (recall Fig. 1). Our design was cross-sectional, so all relationships reported below are associations. We use 'predicted', 'effect size' (f^2), and 'mediation' in their standard statistical senses within the estimated model; these terms carry no causal claims. When we say 'significant', we mean 'statistically significant'.

5.1. RQ1-how/how much: trust $\rightarrow$ usage $\rightarrow$ cognitive engagement?

RQ1-How examines whether and to what extent students' *trust in genAI predicts their routine usage of these tools*, and, in turn, *how that trust and usage relate to their cognitive engagement habits* in STEM coursework—specifically reflection, need for understanding, and critical thinking (recall Sec. 3 for how we model these constructs). We report these findings in a deliberate sequence: we first describe how routine genAI usage relates to cognitive engagement, and then turn to trust.

5.1.1. Usage and cognitive engagement

Fig. 2 shows distributions of cognitive engagement by usage quartile (Q1 = lowest 25%; Q4 = highest 25%).³ Across all three dimensions, median cognitive engagement factor scores declined monotonically from Q1 to Q4. Participants reporting the highest levels of genAI use (Q4) consistently reported the lowest levels of reflection, need for understanding, and critical thinking. Fig. 3 reports the statistical significance of these relationships (visualized as directed paths among

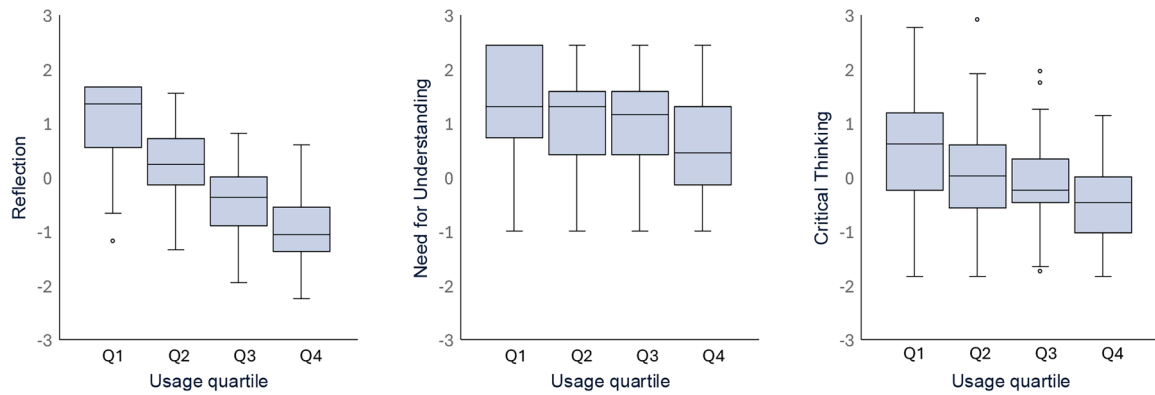


Fig. 2. Cognitive engagement scores by genAI usage. Boxplots show standardized latent scores for Reflection, Need for Understanding, and Critical Thinking across quartiles of reported genAI usage (Q1 = lowest 25%; Q4 = highest 25%). This figure provides a close-up view of data distributions; subsequent figures examine patterns using median-split groups.

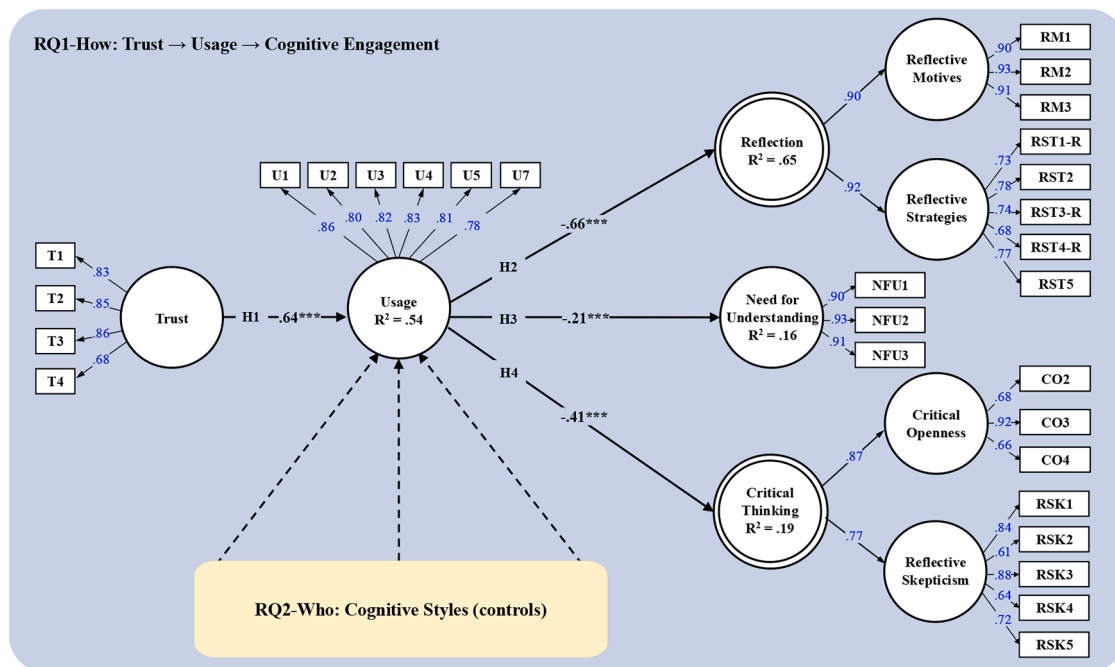


Fig. 3. Structural model results (RQ1-How): Standardized path coefficients (H1–H4) on arrows between constructs (circles) show how Trust and routine genAI Usage relate to Reflection, Need for Understanding, and Critical Thinking, with significance *** $p < .001$, ** $p < .01$, and * $p < .05$. Arrows from constructs to survey items (squares), and from HOCs (double circles) to their lower-order constructs, indicate factor loadings (in blue; all $\geq$ threshold of 0.6 [79]). Dotted arrows and the collapsed yellow region correspond to RQ2-Who results (omitted here for clarity), which are reported in Sec. 5.2.

constructs, e.g., Usage → Reflection is H2). We consider $f^2 < 0.02$ no effect, $f^2 \in [0.02, 0.15]$ small, $f^2 \in [0.15, 0.35]$ medium, and $f^2 > 0.35$ large [93].

(H2a) Usage and reflection: GenAI usage had a significant negative association with reflection ($\beta = -0.66$, $p < .001$), with a large effect size $f^2 = 0.51$, supporting H2a. Specifically, a unit (variance) increase in routine genAI usage predicted a 0.66-unit reduction in reflection, as indicated by the path coefficient (Fig. 3). Reflection requires monitoring oneself—noticing when something does not make sense or when understanding feels superficial [28,29]. The magnitude of this association suggests that routine genAI usage was substantively related to such a process—the pausing, puzzling, and noticing that something requires deeper thought.

³Each score is a model-estimated latent factor score: the measurement model computes how strongly each question signals the underlying construct, then uses these item weights to aggregate the factor score

[57].

(H3a) Usage and need for understanding. GenAI usage had a significant negative association with need for understanding ($\beta = -0.21$, $p < .001$), supporting H3a. The effect size ($f^2 = 0.12$) suggests that while routine genAI use meaningfully predicted need for understanding, stable individual differences (e.g., epistemic curiosity that students bring with them) remain important contributors to this construct. Still, the pattern is notable: participants who routinely used genAI reported significantly less intrinsic drive to grasp why's and how's.

(H4a) Usage and critical thinking. GenAI usage had a significant negative association with critical thinking ($\beta = -0.41$, $p = .001$; $f^2 = 0.20$), supporting H4a. From a dual-process perspective [4], critical thinking involves System 2 work—the deliberate questioning of claims, weighing of alternatives, and scrutiny of assumptions [33,32]. This finding shows that routine genAI use predicted lower levels of such deliberate processing, consistent with the tendency for fluent, seemingly

“complete” genAI outputs to invite System 1 acceptance (as discussed in Sec. 3).

5.1.2. Trust and cognitive engagement

H1 hypothesized that students’ trust in genAI would predict how frequently they used it. For our participants, this turned out to be the case: trust showed a large, significant positive association with routine genAI usage ($\beta = -0.64, p < .001$), with a moderate effect size ($f^2 = 0.25$), supporting H1. This finding is consistent with prior automation-bias accounts [94,21], showing that trust lowers the reluctance towards AI delegation, making it feel reasonable rather than risky or provisional.

Since routine usage was negatively associated with all three dimensions of cognitive engagement (H2a–H4a), we examined the extent to which trust related to the same three dimensions transitively through increased usage (H2b–H4b).

Fig. 4 shows median-split box plot distributions by trust and usage. Across both low- and high-trust groups, participants with higher usage reported lower cognitive engagement (across all three dimensions). Moreover, within each usage group, participants in the “high-trust” group reported even lower cognitive engagement than their “low-trust” peers, indicating that trust amplified these associations.

Mediation analysis confirmed these patterns. Following standard PLS procedures [57], we tested the transitive effects of trust on each cognitive engagement dimension via routine usage. All three effects were significant, supporting H2b–H4b: participants with greater trust in genAI reported significantly less Reflection ($\beta = -0.39, p < .001$,

$f^2 = 0.16$, VAF = 0.89), less Need for Understanding ($\beta = -0.13, p < .001, f^2 = 0.11$, VAF = 0.85), and less Critical thinking ($\beta = -0.26, p = .001, f^2 = 0.15$, VAF = 0.94). All VAF (Variance Accounted For) values were > 0.8 , indicating *full mediation* in the statistical sense [57]: the association between trust and each cognitive engagement dimension was carried entirely by routine genAI usage.

Takeaway: RQ1-How’s findings point to a consistent pattern of cognitive disengagement among students using genAI. Participants *who trusted and routinely used genAI* reported significantly lower levels of *reflection, need for understanding, and critical thinking* in STEM coursework.

5.2. RQ2-who: students’ cognitive styles $\rightarrow$ usage $\rightarrow$ cognitive engagement?

Trust was a strong predictor of routine genAI usage—and, transitively, of cognitive engagement. However, it alone did not fully account

for the observed variance in participants’ routine genAI use ($R^2 = .54$ in Fig. 3). Since individuals differ in how they approach and use technology [38,41,40], here we consider whether the lower cognitive engagement associated with routine usage was concentrated among particular students. If so, which students?

RQ2-Who investigates this question using cognitive styles: whether and how students’ cognitive styles affect their routine use of genAI, and, in turn, their cognitive engagement.

5.2.1. Cognitive styles and routine genAI usage

Prior work has reported certain cognitive styles to significantly influence genAI adoption among knowledge workers [68,42]: intrinsic enjoyment of engaging with technology (*technophilic motivations*), comfort with uncertainty and technology-related risks (*risk tolerance*), and confidence in one’s ability to use such tools effectively (*computer self-efficacy*) [38,39,18,95,40]. Thus, we investigated whether these same cognitive styles also predict *students’* routine usage of genAI.

The results were consistent with prior findings in knowledge work contexts. Participants with stronger technophilic motivations ($\beta = 0.14, p = .011, f^2 = 0.17$), risk tolerance ($\beta = 0.29, p < .001, f^2 = 0.21$), and computer self-efficacy.

($\beta = 0.19, p = .002, f^2 = 0.11$) reported significantly more routine genAI use than their peers (see Fig. 5).

We find this result unsurprising. If a student is motivated by engaging with technology (cognitive style: *technophilic motivations*), it seems likely that they would likewise enjoy engaging with genAI, even when it is known to carry risks (cognitive style: *risk tolerance*), especially if the student is confident that they can handle those risks and use the tool effectively (cognitive style: *computer self-efficacy*). Indeed, these qualities are often touted in STEM education as indicators of career preparedness (e.g. [96–99]). Further, researchers have shown that these qualities *are* often helpful for STEM [39,100]. However, it may be time to question whether this still holds with routine genAI use, which we investigate next.

5.2.2. Cognitive styles and cognitive engagement

If higher technophilic motivations, risk tolerance, and computer self-efficacy remain helpful for STEM education even under conditions of routine genAI use, then students with these cognitive styles should also exhibit higher cognitive engagement in their coursework. For example, among RQ1-How’s findings, the participants in the High Usage/Trust group (recall outliers above the box in Fig. 4) who exhibited fairly high reflection despite heavy genAI use would presumably be those with these cognitive styles.

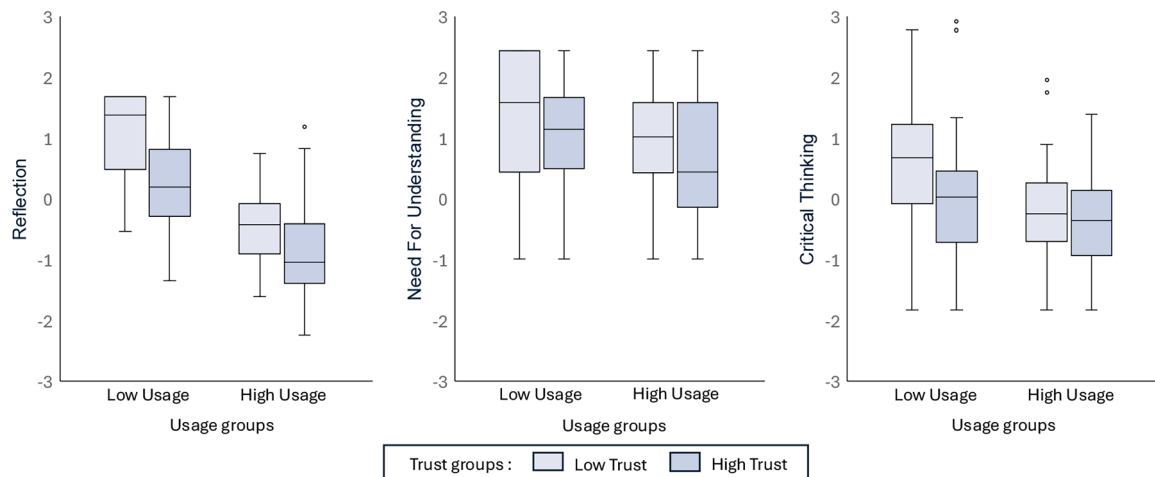


Fig. 4. Trust $\rightarrow$ usage $\rightarrow$ cognitive engagement distributions. Boxplots show standardized latent scores for Reflection, Need for Understanding, and Critical Thinking among participants grouped by genAI usage (median split) and trust in genAI (low vs. high; median split within each usage group).

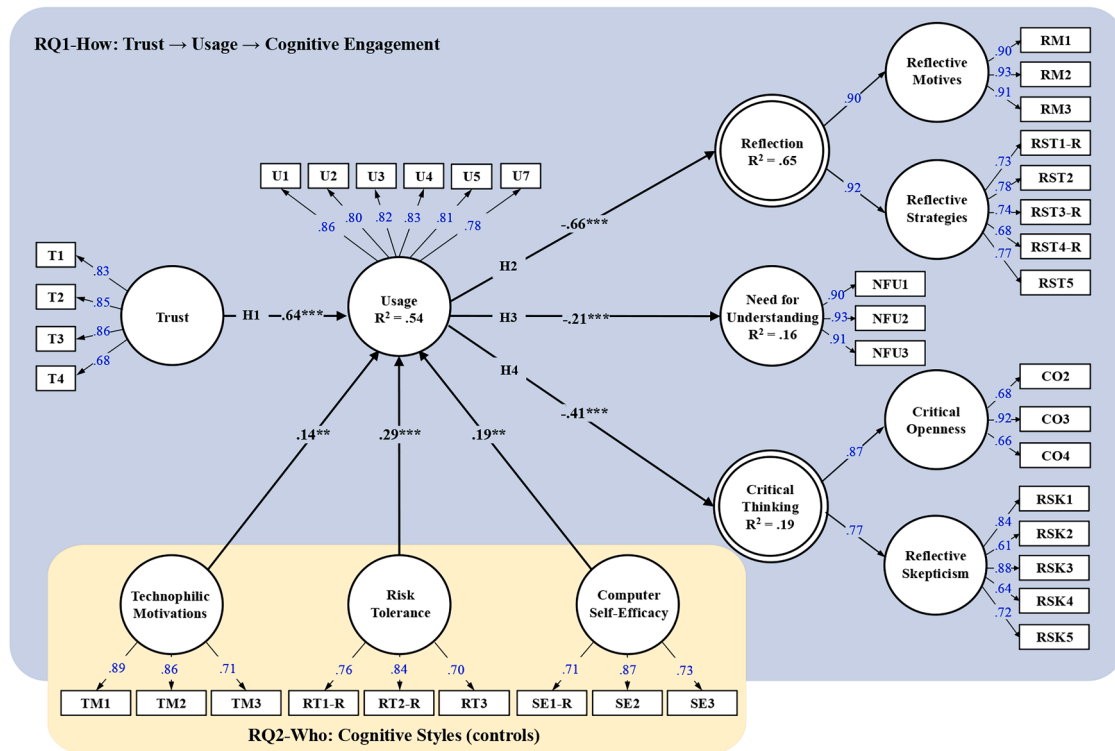


Fig. 5. Structural model path associations (full model: RQs1&2). Standardized path coefficients show direct effects of Trust and cognitive styles (Technophilic Motivations, Risk Tolerance, Computer Self-Efficacy; RQ2—yellow) on Usage, and of Usage on Reflection, Need for Understanding, and Critical Thinking, with significance *** $p < .001$, ** $p < .01$, and * $p < .05$. Paths from LOCs (circles) to items (squares), and from HOCs (double circles) to LOCs, indicate factor loadings (in blue; all ≥ 0.6 [79]).

Technophilic motivations: However, the opposite was true. As Fig. 6 (a) illustrates, higher technophilic motivations corresponded

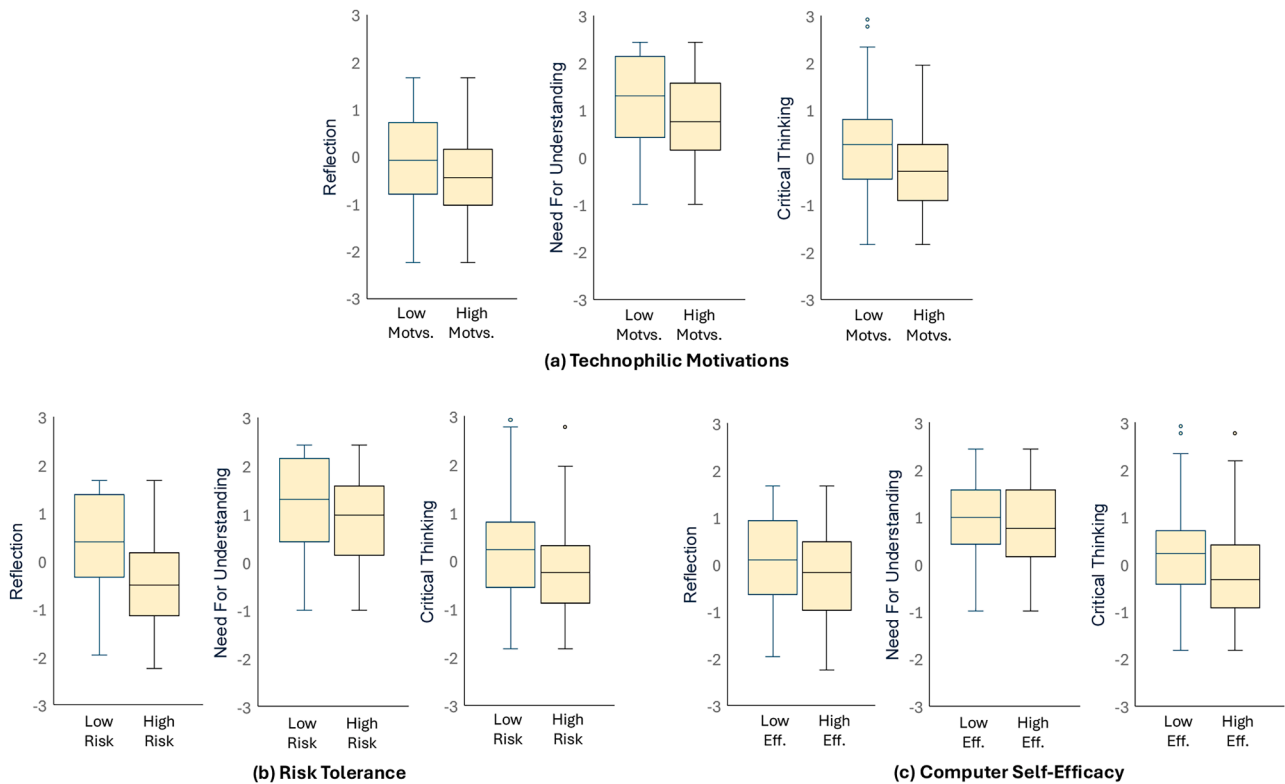


Fig. 6. Cognitive engagement scores by median-split groups on (a) Technophilic Motivations, (b) Risk Tolerance, and (c) Computer Self-Efficacy. Boxplots show standardized latent scores for Reflection, Need for Understanding, and Critical Thinking across all groups.

with lower median cognitive engagement across all three dimensions. Mediation analysis confirmed these patterns: participants with higher technophilic motivations reported significantly less Reflection ($\beta = -0.11, p = .010, f^2 = 0.15, \text{VAF} = 0.83$), significantly less Need for Understanding ($\beta = -0.12, p = .014, f^2 = 0.07, \text{VAF} = 0.83$), and significantly less Critical Thinking ($\beta = -0.05, p = .007, f^2 = 0.09, \text{VAF} = 0.86$).

Risk tolerance and computer self-efficacy: Similar patterns emerged for the other two cognitive styles (Fig. 6 (b) and (c)). Participants with higher risk tolerance reported significantly less Reflection ($\beta = -0.15, p < .001, f^2 = 0.08, \text{VAF} = 0.87$), significantly less Need for Understanding ($\beta = -0.11, p = .002, f^2 = 0.07, \text{VAF} = 0.85$), and significantly less Critical Thinking ($\beta = -0.13, p < .001, f^2 = 0.08, \text{VAF} = 0.82$). Research on attitude towards risk suggests one interpretation why [101]: risk tolerance captures willingness to proceed despite uncertainty. In HAI contexts, this can, alongside increasing usage, reduce users' inclination to question or scrutinize AI outputs [102], even when such scrutiny would be warranted.

Participants with higher computer self-efficacy likewise reported significantly less Reflection ($\beta = 0.17, p = .013, f^2 = 0.15, \text{VAF} = 0.84$), significantly less Need for Understanding ($\beta = 0.12, p = .005, f^2 = 0.07, \text{VAF} = 0.86$), and significantly less Critical Thinking ($\beta = 0.14, p = .020, f^2 = 0.13, \text{VAF} = 0.87$). While prior work suggests that higher self-efficacy leads to greater success (e.g., as per self-efficacy as an effective surrogate for ability research [95,103]), some education research indicates that greater self-confidence in genAI contexts may lead students to overestimate learning gains from such interactions [9,104].

Group analysis: Still, some research suggests that academic level and prior genAI experience may be important factors [72,16]. To investigate whether these factors affected the structural relationships in our model, we conducted standard PLS Multi-Group Analysis (MGA) [57]. We defined groups by (i) academic level (lower- vs. upper-division) and (ii) prior genAI experience (median split on self-reported experience). The analysis revealed no statistically significant differences across any paths, confirming that the model relationships held consistently regardless of participants' academic seniority or prior experience with genAI.

Takeaway: Students that prevailing wisdom would classify as well-positioned for STEM work—those with higher technophilic motivations, risk tolerance, and computer self-efficacy—were also the ones among whom routine genAI use was most strongly associated with lower cognitive engagement.

5.3. Model adequacy assessment

Statistical and practical significance establish that relationships exist, but they do not indicate how well a model accounts for observed variance, whether it is well-specified and consistent with the data (fit), or whether it exhibits predictive relevance beyond the estimation sample. To address these questions, we assessed model adequacy using the coefficient of determination (R^2), model fit (SRMR), and out-of-sample predictive relevance (Q^2), following recommended PLS-SEM practices [57].

R^2 values capture how much variance in each outcome (e.g., Reflection) is explained by its predictors. The model explains a substantial proportion of variance in Usage ($R^2 = 0.54$) and Reflection ($R^2 = 0.65$); our common method bias checks (recall Sec. 4.3) indicated that neither was inflated by method variance. The model explains more moderate variance in Need for Understanding ($R^2 = 0.16$) and Critical Thinking ($R^2 = 0.19$). While these constructs are influenced by a broader set of factors beyond the model, the observed values exceed the common benchmark of 0.15 for exploratory model adequacy in behavioral sciences [57].

Model fit was assessed using the Standardized Root Mean Square Residual (SRMR), which is recommended for detecting structural misspecification in PLS-SEM [57]. The observed SRMR of 0.067 falls below the conservative threshold of 0.08 [105], indicating a well-specified

model, i.e., the proposed relationships are consistent with the covariance structure of the data.

Finally, we assessed out-of-sample predictive relevance to evaluate whether the model generalizes beyond the estimation sample. Using Stone-Geisser's Q^2 [106], computed via 10-fold, 10-repeat PLS-predict, all dependent constructs yielded positive values (Usage: .51, Reflection: .48, Need for understanding: .13, Critical thinking: .27). This indicates that the model's predictions on held-out data outperform a naive mean-based benchmark, supporting its predictive relevance beyond in-sample explanatory fit.

Takeaway: The proposed model is adequate for its intended purpose: it captures sufficient variance in key outcomes, shows no evidence of structural misspecification, and demonstrates predictive relevance beyond in-sample fit.

6. Discussion: the cognitive debt cycle

Our findings raise the possibility of a *cognitive debt cycle* tied to students' trust-driven, routine use of genAI. Much like technical debt accumulates when short-term expedience substitutes for more principled design, cognitive debt may accrue when students repeatedly delegate relevant cognitive work onto genAI. The concern is that genAI may restructure the learning environment in ways that make delegation appear rational, gradually training the brain to reconsider what effort investment feels worthwhile. Further, these effects may be *self-reinforcing*: as students repeatedly delegate cognitive work, their ability to perform these tasks independently may decline. GenAI may then become increasingly necessary, further reducing the already limited opportunities to hone these skills. The result may be a downward spiral of overreliance, in which understanding becomes increasingly shallow, and the capacity to apply knowledge in new or dynamic contexts gradually erodes. Students with weaker engagement habits may also gravitate toward heavier genAI use, and longitudinal designs are needed to disentangle the two.

Unfortunately, the associations we modeled did not weaken with academic seniority or prior genAI experience, challenging the assumption that students will "grow into" reflective users. It is important to note that we captured self-rated genAI experience, which does not indicate how well they used it. Our measures also did not capture whether heavier use was driven by coursework demands. Future work should examine these interactions through measures of interaction quality and course demands. Nevertheless, this raises a fundamental question: *How can students be expected to regulate their genAI use if the very capacities required to do so weaken with routine interactions?*

6.1. Interrupting the cycle: what can educators do?

Recognizing the cognitive debt cycle is one thing; interrupting it is another. A common response has been to call for more AI literacy [107, 108], with the expectation that better-informed learners will be more attuned to the risks of overreliance. Indeed, emerging frameworks, such as the Digital Education Council's AI literacy model [109], position critical thinking and judgment about AI as core competencies, emphasizing the ability to evaluate AI reasoning, identify bias, and recognize contextual gaps in AI outputs. We agree with the importance of these skills. Still, prior work suggests that awareness of AI's fallibility does not reliably deter reliance when answers are immediate and cognitively effortless [21,110]. Our findings echo this pattern: we found no evidence that prior AI experience or academic seniority attenuated the associations, suggesting that *awareness alone may not counteract the incentives to disengage* (subject to the measurement caveats noted above).

Why might this be? Anderson's theory of rational analysis [111] suggests one reason why. According to this theory, humans exhibit a "local maximum of adaptedness" to their environments, optimizing for maximum immediate reward and minimum immediate cost. In educational contexts, this implies that delegating cognitive work to genAI may

be a rational response to environments that reward efficiency while imposing few short-term costs for shallow engagement.

This theory then also suggests possibilities for adjusting students' perceived costs and rewards of AI delegation as a means to interrupt the cognitive debt cycle—affording educators a rethinking of educational experiences where delegating cognitive work becomes less attractive to students.

One possibility centers on increasing the intrinsic rewards of cognitive engagement itself. Tasks that are personally meaningful or inherently enjoyable could reduce the appeal of delegation [112,113], as learners continue to derive satisfaction from the work. For example, designing coursework as opportunities for exploration through storytelling [114], gamified elements [115], or curiosity-driven puzzles (e.g., “Why did the AI get this wrong?” or “How might the answer differ depending on the point of view?” [108]) could tap into intrinsic motivations that invite more deliberate engagement [116]. Notably, curiosity-based approaches may also make the consequences of delegation more visible by surfacing AI errors, omissions, or mismatches within students' own contexts.

Another possibility is to consider increasing the costs of AI delegation through intentional task and activity design. One approach could involve alternating between AI-assisted and independent work. For instance, students might first generate code with AI assistance, then be required to independently critique, debug/repair, verify, and adapt that code to their project requirements without AI (e.g., during a supervised lab). Such designs can make delegation immediately costly, as knowledge gaps that were masked during the AI-assisted phase must be confronted during subsequent independent work. Empirical work supports this sequencing. Students who attempted problems independently before turning to an LLM made greater learning gains [117], and students who read and took notes before consulting an LLM outperformed those who worked with the LLM throughout [14]. Cumulative assignments that build on previous assignments can amplify these costs further, as shallow cognitive engagement in early assignments could compound into visible struggles in later ones. Finally, combining these approaches with in-class assessments and/or collaborative/group activities can further reinforce engagement by leveraging social accountability [118], where shallow AI-generated contributions become quickly visible, potentially compromising individual and/or group reputation and outcomes.

Importantly, these strategies are not “silver bullets” nor miracle cures. As genAI capabilities, human practices, and mindsets co-evolve, responses to this challenge will need to adapt. A key ongoing question for educators remains: *How to design educational experiences to foster cognitive engagement in ways that improve both (1) students' learning of the subject matter and (2) their ability to use AI appropriately?*

6.2. Interrupting the cycle: what can AI designers do?

Interrupting the debt cycle may require more than pedagogical adjustments; it raises questions about how genAI itself can be designed for educational use. One possibility, drawing on Engelbart's early vision of intellectual augmentation [119], is to design genAI as “bicycles for the mind”. Just as bicycles amplify physical efficiency while requiring the rider's active control; genAI, designed as mind-bicycles, could amplify cognitive reach (e.g., by critiquing, counter-arguing, nudging the user [120]) while keeping humans responsible for exploration, steering, and judgment. The goal is for progress to emerge through *complementarity*: the human and the AI each need to contribute distinct forms of cognitive work to achieve meaningful outcomes.

What this framing seeks to avoid is a wholesale substitution of human agency. When genAI “thinks” on a student's behalf, they lose opportunities to build skill, and genAI, in turn, loses access to future high-quality human input shaping its evolution. Designing for complementarity is therefore not only pedagogically desirable; it is necessary for sustaining genAI-assisted knowledge work. A key ongoing challenge

for AI designers then is: *How to design genAI for complementarity in ways that meaningfully amplify human thinking instead of replacing it?*

Incorporating “friction”—intentional points of difficulty [121]—could offer one possibility toward designing such systems for educational use. Designing for friction runs counter to conventional product metrics that prioritize maximal convenience (e.g., fewer clicks). However, prior work shows that friction can be productive when it slows users down to provoke reflection on their emerging work [21,122]. One way to introduce friction, for example, could be to design education-oriented genAI as a “provocateur,” repositioning assistance as questioning rather than confirming [122]. Such a provocateur could prompt students to revisit their assumptions and interrogate genAI's underlying rationales, enabling them to identify and correct flaws in that (and/or their own) reasoning. Another way could be to design systems that require users to deliberately engage with multiple AI perspectives: for instance, by comparing outputs generated under differing viewpoints or competing claims and reconciling disagreements among them. Prior work shows that such designs can indeed strengthen users' critical thinking and reasoning practices [123]. What unites these possibilities is that the friction is deliberate, built into the pedagogy rather than prompted onto a general-purpose tool. Systems designed this way already show gains: an AI tutor engineered around pedagogical best practices such as scaffolding led students to learn more in less time than in-class active learning, while reporting higher engagement [15].

Furthermore, our findings suggest that reimagining genAI for complementarity should not assume a one-size-fits-all approach. Recall, students who might have been regarded as most “tech-ready”—those high in technophilic motivations, self-efficacy, and risk tolerance—were also the ones among whom routine genAI use was most strongly associated with lower reported cognitive engagement (our data speak to how much these students used genAI, not how well, nor why they used it more).

This raises possibilities for adaptivity in education-oriented genAI design to account for students' diverse cognitive styles. For instance, technophilic users may benefit from *interface nudges* that demand comparison and justification (e.g., selecting among multiple AI-generated solutions and justifying their choice, or attempting to “beat the AI”). Similarly, highly self-efficacious users may benefit from scaffolds that require an initial attempt (e.g., plans, assumptions, rationales) before AI assistance is provided, helping recalibrate their expectations from AI support. Encouragingly, systems built along these lines already show gains. For example, an AI tool that adapted its scaffolding to students' cognitive states, e.g., recognizing shallow thinking or disengagement and selecting a matching pedagogical strategy, outperformed a prompt-engineered chatbot baseline on engagement, creativity, and task performance [124]. That said, adaptivity that is too automatic risks undermining user agency and locking students into a fixed interaction style. Although human-AI interaction research is still relatively young, the principle of keeping users in control has been affirmed by many HAI researchers (e.g. [40,125,19,126–128]). Accordingly, adaptivity should remain configurable, enabling students to steer how the system supports them as their contexts and skills evolve.

Ultimately, realizing this future will require coordinated effort among AI designers, researchers, and educators to support students in developing and sustaining essential intellectual habits when using genAI, rather than perpetuating the cognitive debt cycle. While the specific pathways forward remain open and evolving, the cost of inaction is clear: *a future where apparent “efficiency” rises even as sustained thinking habits quietly erode.*

7. Conclusion

We began this investigation by asking: *How do students' trust in and routine use of genAI relate to their cognitive engagement in STEM coursework?* What we found was both clear and concerning. Our main findings were:

RQ1-How: Students who routinely used genAI reported statistically significantly less cognitive engagement, i.e., less reflection, need for understanding, and critical thinking in their coursework. Higher trust in genAI showed a negative indirect association with all three constructs, carried entirely through increased routine genAI use.

RQ2-Who: Ironically, students commonly touted as “tech-natives”—those with stronger technophilic motivations, greater computer self-efficacy, and higher risk tolerance—were the ones among whom routine genAI use was most strongly associated with lower cognitive engagement.

Longitudinal research should investigate whether routine reliance on genAI during formative years changes students’ willingness to engage in effortful thinking, because if it does, many may enter professional life without having developed the intellectual habits that earlier generations developed through practice. Yet this need not be inevitable. The cognitive debt cycle can be interrupted, but doing so will take more than exhortation or awareness campaigns. It calls for deliberate redesign of both learning environments and genAI systems to preserve human agency and support genuine complementarity—a division of cognitive labor in which both human and AI contribute meaningfully, and neither is displaced. The choices educators and AI designers make now will shape not only what students learn, but how they learn to think.

CRedit authorship contribution statement

Rudrajit Choudhuri: Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Christopher A. Sanchez:** Writing – review & editing, Validation, Supervision, Investigation. **Margaret Burnett:** Writing – review & editing, Visualization, Supervision, Funding acquisition. **Anita Sarma:** Writing – review & editing, Validation, Supervision, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Supplementary materials are publicly hosted on [1].

References

- [1] R. Choudhuri, Supplemental package. <https://zenodo.org/record/18420685>, 2026.
- [2] E.F. Risko, S.J. Gilbert, Cognitive offloading, *Trends Cogn. Sci.* 20 (2016) 676–688.
- [3] B. Sparrow, J. Liu, D.M. Wegner, Google effects on memory: cognitive consequences of having information at our fingertips, *Science* 333 (2011) 776–778.
- [4] D. Kahneman, *Thinking, Fast and Slow*, macmillan, 2011.
- [5] H.P. Lee, A. Sarkar, L. Tankelevitch, I. Drosos, S. Rintel, R. Banks, N. Wilson, The impact of generative ai on critical thinking: self-reported reductions in cognitive effort and confidence effects from a survey of knowledge workers, in: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, 2025, pp. 1–22.
- [6] J. Butler, S. Jaffe, R. Janßen, N. Baym, B. Hecht, J. Hofman, S. Rintel, B. Sarrafzadeh, A.M. Sellen, T. Vorvoreanu, J. Teevan, Microsoft New Future of Work Report 2025, Microsoft Research, 2025. Technical Report MSR-TR-2025-58URL, <https://aka.ms/nfw2025>.
- [7] A. Barcaui, Chatgpt as a cognitive crutch: evidence from a randomized controlled trial on knowledge retention, *Soc. Sci. Humanit. Open* 12 (2025) 102287.
- [8] M.T. Chi, R. Wylie, The icap framework: linking cognitive engagement to active learning outcomes, *Educ. Psychol.* 49 (2014) 219–243.
- [9] K. Walker, M. Vorvoreanu, Learning Outcomes with GenAI in the Classroom: a Review of Empirical Evidence, Microsoft Aether Psi Working Group, 2025. Technical Report.
- [10] M. Stadler, M. Bannert, M. Sailer, Cognitive ease at a cost: llms reduce mental effort but compromise depth in student scientific inquiry, *Comput. Hum. Behav.* 160 (2024) 108386.
- [11] Y. Fan, L. Tang, H. Le, K. Shen, S. Tan, Y. Zhao, Y. Shen, X. Li, D. Gašević, Beware of metacognitive laziness: effects of generative artificial intelligence on learning motivation, processes, and performance, *Br. J. Educ. Technol.* 56 (2025) 489–530.
- [12] Kosmyna, N., Hauptmann, E., Yuan, Y.T., Situ, J., Liao, X.H., Beresnitzky, A.V., Braunstein, I., Maes, P., 2025. Your brain on chatgpt: accumulation of cognitive debt when using an ai assistant for essay writing task. *arXiv preprint arXiv: 2506.08872*.
- [13] L. Yan, V. Pammer-Schindler, C. Mills, A. Nguyen, D. Gašević, Beyond efficiency: empirical insights on generative ai’s impact on cognition, metacognition and epistemic agency in learning, *Br. J. Educ. Technol.* 56 (2025) 1675–1685.
- [14] Kreijkes, P., Kewenig, V., Kuvalja, M., Lee, M., Vitello, S., Hofman, J.M., Sellen, A., Rintel, S., Goldstein, D.G., Rothschild, D.M., et al., 2025. Effects of llm use and note-taking on reading comprehension and memory: a randomised experiment in secondary schools. Available at SSRN.
- [15] G. Kestin, K. Miller, A. Kiales, T. Milbourne, G. Ponti, AI tutoring outperforms in-class active learning: an RCT introducing a novel research-based design in an authentic educational setting, *Sci. Rep.* 15 (2025) 17458.
- [16] R. Choudhuri, A. Ramakrishnan, A. Chatterjee, B. Trinkenreich, I. Steinmacher, M. Gerosa, A. Sarma, Insights from the frontline: genAI utilization among software engineering students, in: *2025 IEEE/ACM 37th International Conference on Software Engineering Education and Training (CSEE&T)*, IEEE, 2025, pp. 1–12.
- [17] K. Goddard, A. Roudsari, J.C. Wyatt, Automation bias: a systematic review of frequency, effect mediators, and mitigators, *J. Am. Med. Inform. Assoc.* 19 (2012) 121–127.
- [18] D. Lyell, E. Coiera, Automation bias and verification complexity: a systematic review, *J. Am. Med. Inform. Assoc.* 24 (2017) 423–431.
- [19] R. Choudhuri, D. Liu, I. Steinmacher, M. Gerosa, A. Sarma, How far are we? The triumphs and trials of generative AI in learning software engineering, in: *Proceedings of the IEEE/ACM 46th International Conference on Software Engineering*, 2024, pp. 1–13.
- [20] M.M. Hamid, F.A. Moussaoui, J.N. Guevara, A. Anderson, P. Agarwal, J. Dodge, M. Burnett, Inclusive design of ai’s explanations: just for those previously left out? *ACM Trans. Interact. Intell. Syst.* (2025).
- [21] Z. Bućinca, M.B. Malaya, K.Z. Gajos, To trust or to think: cognitive forcing functions can reduce overreliance on ai in ai-assisted decision-making, *Proc. ACM Hum.-Comput. Interact.* 5 (2021) 1–21.
- [22] X. Chen, K. Ruan, K.P. Ju, N. Yap, X. Wang, More AI assistance reduces cognitive engagement: examining the AI assistance dilemma in AI-supported note-taking, *Proc. ACM Hum.-Comput. Interact.* 9 (2025) 1–29.
- [23] Dewey, J., 1933. How we think: a restatement of the relation of reflective thinking to the educative process.
- [24] J. Biggs, D. Kember, D.Y. Leung, The revised two-factor study process questionnaire: R-spq-2f, *Br. J. Educ. Psychol.* 71 (2001) 133–149.
- [25] P.R. Pintrich, E.V. De Groot, Motivational and self-regulated learning components of classroom academic performance, *J. Educ. Psychol.* 82 (1990) 33.
- [26] D. Kember, D.Y. Leung, A. Jones, A.Y. Loke, J. McKay, K. Sinclair, H. Tse, C. Webb, F.K. Yuet Wong, M. Wong, et al., Development of a questionnaire to measure the level of reflective thinking, *Assess. Eval. High. Educ.* 25 (2000) 381–395.
- [27] W. Li, J.Y. Huang, C.Y. Liu, J.C. Tseng, S.P. Wang, A study on the relationship between student learning engagements and higher-order thinking skills in programming learning, *Think. Ski. Creat.* 49 (2023) 101369.
- [28] J. Dunlosky, C. Hertzog, M. Kennedy, K.W. Thiede, The self-monitoring approach for effective learning, *Cogn. Technol.* 10 (2005) 4–11.
- [29] N. Silver, M. Kaplan, D. LaVaque-Manty, D. Meizlish, Using Reflection and Metacognition to Improve Student Learning: Across the Disciplines, Across the Academy, Taylor & Francis, 2023.
- [30] M.T. Chi, Active-constructive-interactive: a conceptual framework for differentiating learning activities, *Top. Cogn. Sci.* 1 (2009) 73–105.
- [31] J.T. Cacioppo, R.E. Petty, The need for cognition, *J. Personal. Soc. Psychol.* 42 (1982) 116.
- [32] Facione, P., 1990. Critical thinking: a statement of expert consensus for purposes of educational assessment and instruction (the delphi report).
- [33] E.M. Sosu, The development and psychometric validation of a critical thinking disposition scale, *Think. Ski. Creat.* 9 (2013) 107–119.
- [34] P.A. Facione, et al., Critical thinking: what it is and why it counts, *Insight Assess.* 1 (2011) 1–23.
- [35] M. Boekaerts, M. Zeidner, P.R. Pintrich, *Handbook of Self-Regulation*, Elsevier, 1999.

- [36] C.J. Evans, J.R. Kirby, L.R. Fabrigar, Approaches to learning, need for cognition, and strategic flexibility among university students, *Br. J. Educ. Psychol.* 73 (2003) 507–528.
- [37] R.J. Sternberg, E.L. Grigorenko, Are cognitive styles still in style? *Am. Psychol.* 52 (1997) 700.
- [38] M. Burnett, S. Stumpf, J. Macbeth, S. Makri, L. Beckwith, I. Kwan, A. Peters, W. Jernigan, Gendermag: a method for evaluating software's gender inclusiveness, *Interact. Comput.* 28 (2016) 760–787.
- [39] M. Martínez-Córcoles, M. Teichmann, M. Murdvee, Assessing technophobia and technophilia: development and validation of a questionnaire, *Technol. Soc.* 51 (2017) 183–188.
- [40] A. Anderson, J.N. Guevara, F. Moussaoui, T. Li, M. Vorvoreanu, M. Burnett, Measuring user experience inclusivity in human-ai interaction via five user problem-solving styles, *ACM Trans. Interact. Intell. Syst.* 14 (2024) 1–90.
- [41] M. Vorvoreanu, L. Zhang, Y.H. Huang, C. Hilderbrand, Z. Steine-Hanson, M. Burnett, From gender biases to gender-inclusive design: an empirical investigation, in: *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 2019, pp. 1–14.
- [42] Choudhuri, R., Trinkenreich, B., Pandita, R., Kalliamvakou, E., Steinmacher, I., Gerosa, M., Sanchez, C., Sarma, A., 2025. What needs attention? prioritizing drivers of developers' trust and adoption of generative ai. *arXiv preprint arXiv: 2505.17418*.
- [43] M. Gerlich, Ai tools in society: impacts on cognitive offloading and the future of critical thinking, *Societies* 15 (2025) 6.
- [44] J. Prather, J. Leinonen, N. Kiesler, J. Gerson Benario, S. Lau, S. MacNeil, N. Norouzi, S. Opel, V. Pettit, L. Porter, et al., Beyond the hype: a comprehensive review of current trends in generative ai research, teaching practices, and tools. 2024 Working Group Reports on Innovation and Technology in Computer Science Education, 2025, pp. 300–338.
- [45] Qin, Q., Santos, R.d.S., Spinola, R., 2025. On the role and impact of genai tools in software engineering education. *arXiv preprint arXiv:2512.04256*.
- [46] J. Wang, W. Fan, The effect of chatgpt on students' learning performance, learning perception, and higher-order thinking: insights from a meta-analysis, *Humanit. Soc. Sci. Commun.* 12 (2025).
- [47] A. Nguyen, Y. Hong, B. Dang, X. Huang, Human-ai collaboration patterns in ai-assisted academic writing, *Stud. High. Educ.* 49 (2024) 847–864.
- [48] L. Zhang, J. Xu, The paradox of self-efficacy and technological dependence: unraveling generative ai's impact on university students' task completion, *Internet High. Educ.* 65 (2025) 100978.
- [49] J. Prather, B.N. Reeves, J. Leinonen, S. MacNeil, A.S. Randrianasolo, B.A. Becker, B. Kimmel, J. Wright, B. Briggs, The widening gap: the benefits and harms of generative ai for novice programmers, in: *Proceedings of the 2024 ACM Conference on International Computing Education Research* 1, 2024, pp. 469–486.
- [50] J. Prather, B.N. Reeves, P. Denny, B.A. Becker, J. Leinonen, A. Luxton-Reilly, G. Powell, J. Finnie-Ansley, E.A. Santos, It's weird that it knows what i want": usability and interactions with copilot for novice programmers, *ACM Trans. Comput.-Hum. Interact.* 31 (2023) 1–31.
- [51] M. Lepp, J. Kaimre, Does generative ai help in learning programming: students' perceptions, reported use and relation to performance, *Comput. Hum. Behav. Rep.* 18 (2025) 100642.
- [52] Miklian, J., Hoelscher, K., 2025. A new digital divide? coder worldviews, the slop economy, and democracy in the age of AI. *arXiv preprint arXiv:2510.04755*.
- [53] L.E. Margulieux, J. Prather, B.N. Reeves, B.A. Becker, G. Cetin Uzun, D. Loksa, J. Leinonen, P. Denny, Self-regulation, self-efficacy, and fear of failure interactions with how novices use llms to solve programming problems, in: *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V 1*, 2024, pp. 276–282.
- [54] J.A. Fredricks, P.C. Blumenfeld, A.H. Paris, School engagement: potential of the concept, state of the evidence, *Rev. Educ. Res.* 74 (2004) 59–109.
- [55] S. Lamborn, F. Newmann, G. Wehlage, The significance and sources of student engagement. *Student Engagement and Achievement in American Secondary Schools*, 1992, pp. 11–39.
- [56] T. Raykov, G.A. Marcoulides, *Introduction to Psychometric Theory*, Routledge, 2011.
- [57] J.F. Hair, A Primer on Partial Least Squares Structural Equation Modeling (PLS-SEM), Sage, 2014.
- [58] R.H. Ennis, Critical thinking assessment, *Theory Pract.* 32 (1993) 179–186.
- [59] R.H. Ennis, Critical thinking across the curriculum: a vision, *Topoi* 37 (2018) 165–184.
- [60] P.A. Facione, C.A. Sanchez, N.C. Facione, J. Gainen, The disposition toward critical thinking, *J. Gen. Educ.* 44 (1995) 1–25.
- [61] B.S. Bloom, M.D. Engelhart, E.J. Furst, W.H. Hill, D.R. Krathwohl, et al., *Taxonomy of Educational Objectives: the Classification of Educational Goals. Handbook 1: Cognitive Domain*, Longman New York, 1956.
- [62] Paul, R.W., Elder, L., Bartell, T., 1997. California teacher preparation for instruction in critical thinking: research findings and policy recommendations.
- [63] J.D. Lee, K.A. See, Trust in automation: designing for appropriate reliance, *Hum. Factors* 46 (2004) 50–80.
- [64] Q.V. Liao, S.S. Sundar, Designing for responsible trust in AI systems: a communication perspective, in: *2022 ACM Conference on Fairness, Accountability, and Transparency*, 2022, pp. 1257–1268.
- [65] O. Vereschak, G. Bailly, B. Caramiaux, How to evaluate trust in AI-assisted decision making? a survey of empirical methodologies, *Hum.-Comput. Interact.* 5 (2021) 1–39.
- [66] Wang, R., Cheng, R., Ford, D., Zimmermann, T., 2023. Investigating and designing for trust in AI-powered code generation tools. *arXiv preprint arXiv: 2305.11248*.
- [67] M. Wischniewski, N. Krämer, E. Müller, Measuring and understanding trust calibrations for automated systems: a survey of the state-of-the-art and future directions, in: *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, 2023, pp. 1–16.
- [68] R. Choudhuri, B. Trinkenreich, R. Pandita, E. Kalliamvakou, I. Steinmacher, M. Gerosa, C. Sanchez, A. Sarma, What guides our choices? Modeling developers' trust and behavioral intentions towards GenAI, in: *2025 IEEE/ACM 47th International Conference on Software Engineering (ICSE)*, 2025, <https://doi.org/10.1109/ICSE55347.2025.00087>.
- [69] S.A. Perrig, N. Scharowski, F. Brühlmann, Trust issues with trust scales: examining the psychometric quality of trust measures in the context of AI, in: *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*, 2023, pp. 1–7.
- [70] M. Abbas, F.A. Jam, T.I. Khan, Is it harmful or helpful? examining the causes and consequences of generative ai usage among university students, *Int. J. Educ. Technol. High. Educ.* 21 (2024) 10.
- [71] M.M. Hamid, A. Chatterjee, M. Guizani, A. Anderson, F. Moussaoui, S. Yang, I. Escobar, A. Sarma, M. Burnett, How to measure diversity actionably in technology. Equity, Diversity, and Inclusion in Software Engineering: Best Practices and Insights, Apress, Berkeley, CA, 2024, pp. 469–485.
- [72] K.E. Singer-Freeman, K. Verbeke, B. Barre, Generative ai use among university students depends on academic level and task, *High. Learn. Res. Commun.* 15 (2025) 8.
- [73] B.A. Kitchenham, S.L. Pfleeger, Personal opinion surveys. *Guide to Advanced Empirical Software Engineering*, Springer, 2008, pp. 63–92.
- [74] Qualtrics, Qualtrics survey platform. <https://www.qualtrics.com>, 2025. Accessed August 11, 2025.
- [75] W.L. Grichting, The meaning of "i don't know" in opinion surveys: indifference versus ignorance, *Aust. Psychol.* 29 (1994).
- [76] F. Faul, E. Erdfelder, A. Buchner, A.G. Lang, Statistical power analyses using g* power 3.1: tests for correlation and regression analyses, *Behav. Res. Methods* 41 (2009) 1149–1160.
- [77] D. Russo, Navigating the complexity of generative AI adoption in software engineering, *ACM Trans. Softw. Eng. Methodol.* (2024).
- [78] M. Sarstedt, C.M. Ringle, J.F. Hair, Treating unobserved heterogeneity in PLS-SEM: a multi-method approach. *Partial Least Squares Path Modeling*, Springer, 2017, pp. 197–217.
- [79] J.F. Hair, J.J. Risher, M. Sarstedt, C.M. Ringle, When to use and how to report the results of PLS-SEM, *Eur. Bus. Rev.* (2019).
- [80] SmartPLS Team, Smartpls (version 4.1.0) [computer software], URL, <http://smartpls.com/>, 2024. accessed: July 7, 2024.
- [81] J.M. Becker, K. Klein, M. Wetzels, Hierarchical latent variable models in pls-sem: guidelines for using reflective-formative type models, *Long Range Plan.* 45 (2012) 359–394.
- [82] M. Sarstedt, J.F. Hair, J.H. Cheah, et al., How to specify, estimate, and validate higher-order constructs in pls-sem, *Australas. Mark. J.* 27 (2019) 197–211.
- [83] M.C. Howard, A review of exploratory factor analysis decisions and overview of current practices: what we are doing and how can we improve? *Int. J. Hum.-Comput. Interact.* 32 (2016) 51–62.
- [84] N. Kock, Advanced mediating effects tests, multi-group analyses, and measurement model assessments in PLS-based SEM, *Int. J. E-Collab.* 10 (2014).
- [85] J. Henseler, C.M. Ringle, M. Sarstedt, A new criterion for assessing discriminant validity in variance-based structural equation modeling, *J. Acad. Mark. Sci.* 43 (2015) 115–135.
- [86] P.M. Podsakoff, S.B. MacKenzie, J.Y. Lee, N.P. Podsakoff, Common method biases in behavioral research: a critical review of the literature and recommended remedies, *J. Appl. Psychol.* 88 (2003) 879.
- [87] N. Kock, Common method bias in pls-sem: a full collinearity assessment approach, *Int. J. E-Collab. (IJEC)* 11 (2015) 1–10.
- [88] Sarstedt, M., Cheah, J.H., 2019. Partial least squares structural equation modeling using smartPLS: a software review.
- [89] A.J. Ko, T.D. LaToza, M.M. Burnett, A practical guide to controlled experiments of software engineering tools with human participants, *Empir. Softw. Eng.* 20 (2015) 110–141.
- [90] K.J. Stol, B. Fitzgerald, The ABC of software engineering research, *ACM TOSEM* 27 (2018).
- [91] M. Amoozadeh, D. Daniels, D. Nam, A. Kumar, S. Chen, M. Hilton, S. Srinivasa Ragavan, M.A. Alipour, Trust in generative ai among students: an exploratory study, in: *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V 1*, 2024, pp. 67–73.
- [92] F. Shull, J. Singer, D.I. Sjøberg, *Guide to Advanced Empirical Software Engineering*, 2007.
- [93] J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, Routledge, 2013.
- [94] R. Parasuraman, V. Riley, Humans and automation: use, misuse, disuse, abuse, *Hum. Factors* 39 (1997) 230–253.
- [95] A. Bandura, *Self-Efficacy: the Exercise of Control*, Macmillan, 1997.
- [96] R. Spair, It ain't your father's tech employer anymore: the shift in job security and workplace culture, URL, <https://www.linkedin.com/pulse/aint-your-fathers-tech-employer-anymore-shift-job-security-rick-spair-hw3ze/>, 2025. accessed: January 7, 2026.
- [97] Parents Canada Sponsored Content, Getting students excited about stem requires taking risks, URL, <https://parentscanada.com/sponsored/getting-students-excited-about-stem/>, 2022. accessed: January 8, 2026.

- [98] Stringer, K., 2016. Finding success in failure: stem educators say student risk-taking is key to real-world learning. URL: <https://www.the74million.org/article/finding-success-in-failure-stem-educators-say-student-risk-taking-is-key-to-real-world-accessed>: January 8, 2026.
- [99] C. Sakellariou, Z. Fang, Self-efficacy and interest in stem subjects as predictors of the stem gender gap in the us: the role of unobserved heterogeneity, *Int. J. Educ. Res.* 109 (2021) 101821.
- [100] S.C. Giaccone, M. Magnusson, Unveiling the role of risk-taking in innovation: antecedents and effects, *R&D Manag.* 52 (2022) 93–107.
- [101] G.F. Loewenstein, E.U. Weber, C.K. Hsee, N. Welch, Risk as feelings, *Psychol. Bull.* 127 (2001) 267.
- [102] C. Zhai, S. Wibowo, L.D. Li, The effects of over-reliance on ai dialogue systems on students' cognitive abilities: a systematic review, *Smart Learn. Environ.* 11 (2024) 28.
- [103] D.R. Compeau, C.A. Higgins, Computer self-efficacy: development of a measure and initial test, *MIS Q.* (1995) 189–211.
- [104] L. Fan, K. Deng, F. Liu, Educational impacts of generative artificial intelligence on learning and performance of engineering students in China, *Sci. Rep.* 15 (2025) 26521.
- [105] J. Henseler, G. Hubona, P.A. Ray, Using pls path modeling in new technology research: updated guidelines, *Ind. Manag. Data Syst.* 116 (2016) 2–20.
- [106] M. Stone, Cross-validated choice and assessment of statistical predictions, *J. R. Stat. Soc. B Stat. Methodol.* (1974) 36.
- [107] X. Gu, B.J. Ericson, Ai literacy in k-12 and higher education in the wake of generative ai: an integrative review, in: *Proceedings of the 2025 ACM Conference on International Computing Education Research V 1*, 2025, pp. 125–140.
- [108] E. Kasneci, K. Seßler, S. Küchemann, M. Bannert, D. Dementieva, F. Fischer, U. Gasser, G. Groh, S. Günemann, E. Hüllermeier, et al., Chatgpt for good? on opportunities and challenges of large language models for education, *Learn. Individ. Differ.* 103 (2023) 102274.
- [109] Digital Education Council, Digital education council ai literacy framework. <https://www.digitaleducationcouncil.com/post/digital-education-council-ai-literacy-framework>, 2025. Accessed December 11, 2025.
- [110] B.W. Stone, Generative ai in higher education: uncertain students, ambiguous use cases, and mercenary perspectives, *Teach. Psychol.* (2024).
- [111] J.R. Anderson, *The Adaptive Character of Thought*, Psychology Press, 2013.
- [112] R.S. Lazarus, *Emotion and Adaptation*, Oxford University Press, 1991.
- [113] Roseman, I.J., Smith, C.A., 2001. Appraisal theory. appraisal processes in emotion: theory, methods, research, 3–19.
- [114] R.C. Clark, R.E. Mayer, *E-learning and the Science of Instruction: Proven Guidelines for Consumers and Designers of Multimedia Learning*, John Wiley & Sons, 2023.
- [115] J. Hamari, J. Koivisto, H. Sarsa, Does gamification work? A literature review of empirical studies on gamification, in: *2014 47th Hawaii International Conference on System Sciences*, IEEE, 2014, pp. 3025–3034.
- [116] A. Costantini, A.B. Bakker, Y.S. Scharp, Playful study design: a novel approach to enhancing student well-being and academic performance, *Educ. Psychol. Rev.* 37 (2025) 1–42.
- [117] Kumar, H., Rothschild, D.M., Goldstein, D.G., Hofman, J.M., 2023. Math education with large language models: peril or promise? Available at SSRN 4641653.
- [118] K.R. Wentzel, Social competence at school: relation between social responsibility and academic achievement, *Rev. Educ. Res.* 61 (1991) 1–24.
- [119] D.C. Engelbart, Augmenting human intellect: a conceptual framework. *Augmented Education in the Global Age*, Routledge, 2023, pp. 13–29.
- [120] L. Reicherts, Z.T. Zhang, E. von Oswald, Y. Liu, Y. Rogers, M. Hassib, Ai, help me think—but for myself: assisting people in complex decision-making by providing different kinds of cognitive support, in: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, 2025, pp. 1–19.
- [121] A.L. Cox, S.J. Gould, M.E. Cecchinato, I. Iacovides, I. Renfree, Design frictions for mindful interactions: the case for microboundaries, in: *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems*, 2016, pp. 1389–1397.
- [122] Drosos, I., Sarkar, A., Toronto, N., et al., 2025. "It makes you think": provocations help restore critical thinking to ai-assisted knowledge work. *arXiv preprint arXiv:2501.17247*.
- [123] Liu, Y., Shah, V., Suh, S., Siangliulue, P., August, T., Huang, Y., 2025. Perspectra: choosing your experts enhances critical thinking in multi-agent research ideation. *arXiv preprint arXiv:2509.20553*.
- [124] S. Zha, Y. Liu, C. Zheng, J. Xu, F. Yu, J. Gong, Y. Xu, Mentigo: an intelligent agent for mentoring students in the creative problem solving process, in: *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, 2025, pp. 1–22.
- [125] S. Amershi, D. Weld, M. Vorvoreanu, A. Fournery, B. Nushi, P. Collisson, J. Suh, S. Iqbal, P.N. Bennett, K. Inkpen, et al., Guidelines for human-AI interaction, in: *2019 CHI Conference on Human Factors in Computing Systems*, 2019, pp. 1–13.
- [126] P. Palanque, F. Paternò, V. Roto, A. Schmidt, S. Stumpf, J. Ziegler, A multi-perspective panel on user-centred transparency, explainability, and controllability in automations, in: *IFIP Conference on Human-Computer Interaction*, Springer, 2023, pp. 349–353.
- [127] B. Shneiderman, *Human-Centered AI*, Oxford University Press, 2022.
- [128] A. Schmidt, Interactive human centered artificial intelligence: a definition and research challenges, in: *Proceedings of the 2020 International Conference on Advanced Visual Interfaces*, 2020, pp. 1–4.